

100 Days of Azure DevOps

Day 22 Handout — Pick the agent like a capacity decision, not a personality trait

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — hosted vs self-hosted (same YAML job, different product)

Dimension	Microsoft-hosted	Self-hosted
What it is	Microsoft-provided VM image for the job	A machine / scale set you operate
YAML	pool: vmlImage: ubuntu-latest	pool: Default (named pool)
Images	ubuntu-latest, windows-latest, macOS SKUs with SDKs on the image	Whatever you installed. You own drift.
Patching / disk	Microsoft patches the VM. Workspace dies with the job	You patch OS, agents, AV, certs. Disk fills with leftover workspaces
Network	Public Microsoft IPs. No hop into your VNet unless you design for it	Can sit on a VNet / on-prem. Use when the job must reach private endpoints
Licensed tools	Public SDKs only. No ISV compiler that cannot leave the building	Install the licensed compiler, private NuGet, air-gapped feed
Failure mode	Queue delay / hosted outage / image SKU change	Agent offline, capability mismatch, scale-set at 0, 2am patch window
Bill	Microsoft-hosted minutes + parallel-job SKU	Parallel-job SKU PLUS the VM / scale set. Two bills, not one

Architecture B — decide with a constraint, not a blog title

Constraint	Default	Move off hosted when
Public SaaS build	Hosted ubuntu-latest	Never, unless a later constraint appears
Private network hop	Hosted cannot see it	Job must hit a VNet-only feed, private AKS API, or on-prem artifact share
Licensed / air-gapped tool	Not on the Microsoft image	Compiler, scanner, or SDK that policy forbids on a public VM
Mac / Windows SKU need	Use the hosted image first	Image lacks the SDK pin you actually build with — then pin or self-host
Control as a feeling	Still hosted	We want control is not a constraint. Offline agents at 2am is the cost of that feeling

Architecture C — pitfalls that queue jobs forever

Pitfall	What you see	Fix
Capability folklore	Job waits on a pool named the Java agent	Demand a registered capability in YAML, not a nickname in Slack
Pool sprawl	Five pools, nobody knows which one is live	One pool per real constraint (OS / network / license). Delete the rest
Missing parallel job	Self-hosted VM is up; pipeline still queued	Self-hosted still needs an Azure DevOps parallel job. The VM bill is extra
Agent version drift	Worked Friday, fails Monday after an agent update	Pin agent version on scale sets. Treat agents as cattle with a known image
Workspace leftovers	Disk 100%, jobs fail with no obvious YAML error	Clean workspaces; prefer scale-set agents that die with the job

One-liner to remember

Hosted is Uber. Self-hosted is owning the car — insurance, parking, and the 2am flat tyre included.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps org only. Project azure-100-labs (or today's lab project).

- 1 Open Microsoft Learn: hosted vs self-hosted agents. Write three constraints that would force self-hosted (private network, licensed tool, policy). If you cannot name one, hosted stays the default.
- 2 Org settings → Agent pools. List every pool. Note Microsoft-hosted vs any Default / self-hosted pool. Screenshot nothing from an employer org.
- 3 Open yesterday's YAML (Day 21). Confirm pool: vmImage: ubuntu-latest. That is the rule for every remaining 100-day lab unless a private-network requirement appears.
- 4 In the last pipeline run, open the job log. Find the image name / hosted agent line. That receipt is how you prove hosted actually ran.
- 5 Write docs/agents-day22.md with: (1) the three constraints, (2) the pool list from this org, (3) the one-sentence rule you will follow for Days 23–100.
- 6 Do not install the agent on your laptop 'for practice'. That creates a 2am offline pool you now own.

Starter YAML — hosted stays the default

```
trigger:
  - main
pool:
  vmImage: ubuntu-latest # Microsoft-hosted. Do not switch to a named pool for labs.
steps:
  - script: echo Agent image is $(Agent.OS) / $(Agent.Name)
    displayName: Prove which agent ran
```

Done checklist

- Can explain hosted vs self-hosted without saying we wanted control
- Named at least one real constraint that would justify self-hosted — or wrote none yet
- YAML still uses vmImage: ubuntu-latest
- docs/agents-day22.md exists on the personal repo

Tomorrow — Day 23

YAML pipeline basics